Rafael Gonçalves

GHC 9223, Carnegie Mellon University

rgoncalv@andrew.cmu.edu | rafa1906.github.io | rafa1906 | rafa1906

Profile & Research Interests

Broadly, I am interested in **software security** and **formal methods**. I am particularly interested in static and dynamic program analysis, and in developing tools that allow developers to efficiently analyze real-world software.

Education

Carnegie Mellon University & Instituto Superior Técnico, University of Lisbon *Pittsburgh, PA & Lisbon, Portugal*
PHD IN COMPUTER SCIENCE *Aug. 2024 - present*
- Advised by Prof. José Fragoso Santos, Prof. Limin Jia & Prof. Pedro Adão
- Supported by a CMU Portugal Dual Degree PhD fellowship

Instituto Superior Técnico, University of Lisbon *Lisbon, Portugal*
MSC IN COMPUTER SCIENCE & ENGINEERING *Sep. 2021 - Nov. 2023*
- **GPA:** 4.0/4.0 (PT: 18/20)
- **Specializations:** Cybersecurity · Distributed Systems
- **Master Thesis:** Specification-Driven Synthesis of Summaries for Symbolic Execution (advised by Prof. José Fragoso Santos & Prof. Pedro Adão)

Instituto Superior Técnico, University of Lisbon *Lisbon, Portugal*
BSC IN COMPUTER SCIENCE & ENGINEERING *Sep. 2018 - Jul. 2021*
- **GPA:** 3.883/4.0 (PT: 17/20)

Experience

INESC-ID *Lisbon, Portugal*
RESEARCH ASSISTANT *Jan. 2024 - Oct. 2024*
- **Group:** Automated Reasoning and Software Reliability (ARSR)
- **Project:** "RIGA: Reasoning Over Indirect Discrimination" (DOI: 10.54499/2022.03537.PTDC)
- **Supervisor:** Prof. José Fragoso Santos

Carnegie Mellon University *Pittsburgh, PA*
VISITING SCHOLAR *Oct. 2023 - Dec. 2023*
- Worked on extending NODEMEDIC, a dynamic taint analysis tool, to detect prototype pollution vulnerabilities
- **Host:** Prof. Limin Jia

Publications

R. Gonçalves, F. Ramos, P. Adão, J. Fragoso Santos. Specification-Driven Generation of Summaries for Symbolic Execution. To appear in: European Symposium on Programming (ESOP 2026), Turin, Italy, 2026.

R. Gonçalves, F. Gouveia, I. Lynce, J. Fragoso Santos. Proxy Attribute Discovery in Machine Learning Datasets via Inductive Logic Programming. In: Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2025), Hamilton, Canada, 2025.

R. Gonçalves, F. Ramos, P. Adão, J. Fragoso Santos. Poster: Specification-Driven Synthesis of Summaries for Symbolic Execution. Presented at: ISSTA/ECOOP 2023, Seattle, WA, USA, 2023.

Teaching

Spring 2023, 2024	**Highly Dependable Systems (MSc)**, Teaching Assistant
Summer 2024	**Compilers (BSc)**, Teaching Assistant

Courses

Instituto Superior Técnico, University of Lisbon *Lisbon, Portugal*

BINARY ANALYSIS WITH APPLICATIONS TO MACHINE AND DEEP LEARNING *Feb. 2023 - Mar. 2023*

- Taught by: Prof. Michele Ianni (University of Calabria)

Honors & Awards

2024-29	**Dual Degree PhD Fellowship**, CMU Portugal
2024	**Outstanding Teaching Award**, Instituto Superior Técnico, University of Lisbon
2021	**Academic Excellence Diploma**, Instituto Superior Técnico, University of Lisbon
2018-20, 2023	**Academic Merit Diploma**, Instituto Superior Técnico, University of Lisbon

Other Skills

Programming	C · Python · Haskell · JavaScript · Java · Rust
Languages	Portuguese (native) · English (fluent) · French (basic)